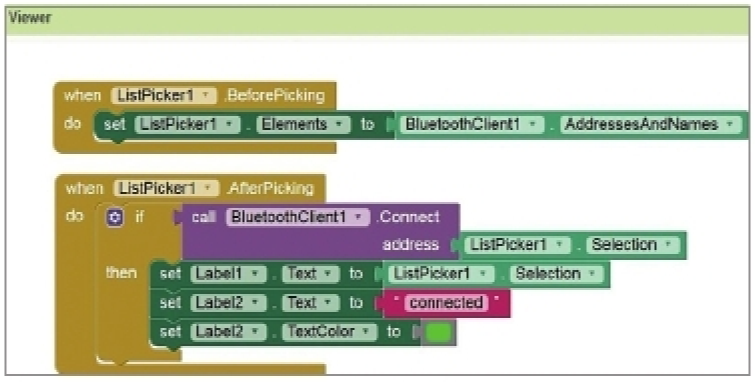

Figure 17: Blocks programming for ListPicker

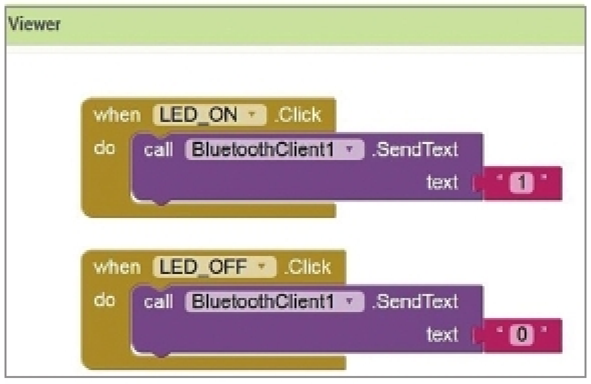

Figure 18: Blocks programming for LED

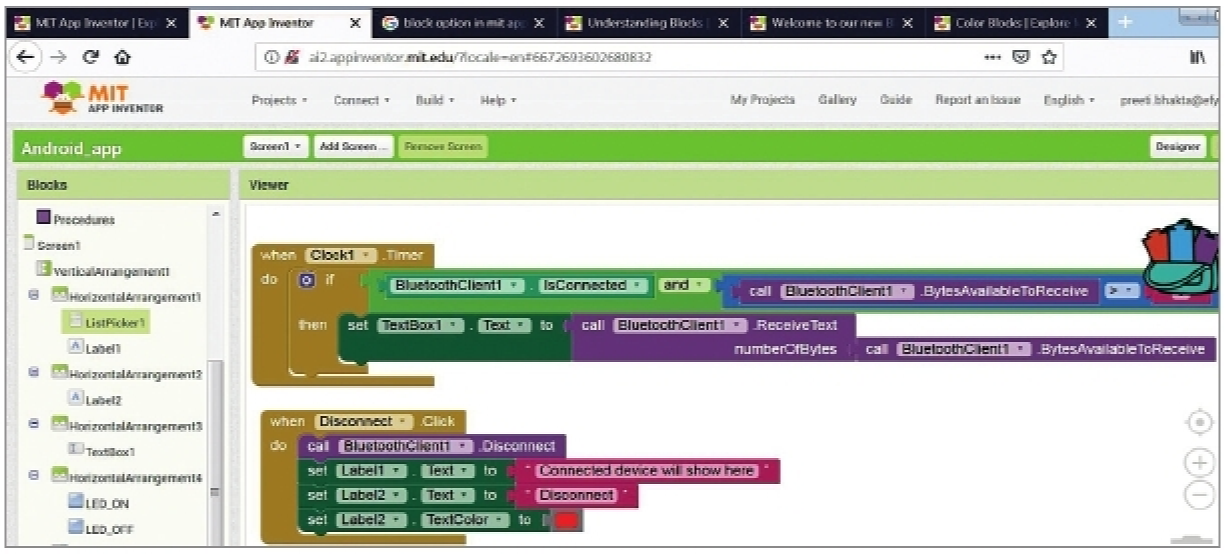

Figure 19: Blocks programming for clock and disconnect functions

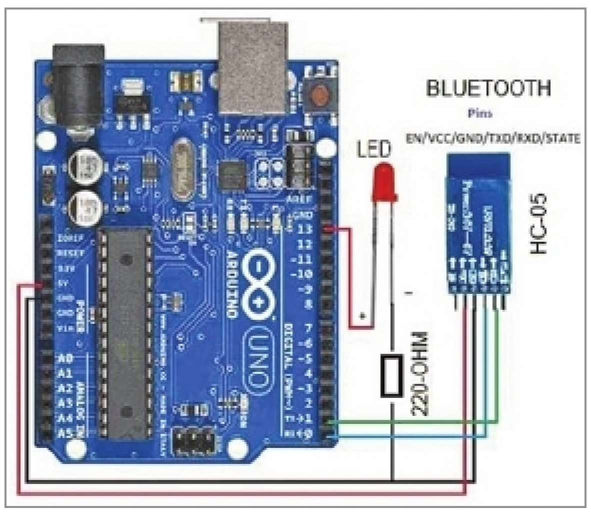

Figure 20: Circuit diagram of Arduino with Bluetooth module

Select BytesAvailableToReceive function of BluetoothClient1, which is greater than 0. For attaching 0, select a text string and enter 0. Here, 0 means bytes are available.

If the 'if' condition is true, convert Bytes Available on Bluetooth into text and set TextBox1.Text as the available text.

Adding Disconnect: Select Disconnect from Blocks. When it is clicked, call a function of BluetoothClient1.Disconnect, and set label1 to Connected Device Will Show Here and label2 to Disconnected. And, change colour to red, as shown in Figure 19. With this, the application designing process is completed.

(***Note.*** You can also refer to 'How To Make Android App' doc file included alongwith the source code at *source.efymag.com*)

Generating Android application file

The next step is to check the above app on an Android smartphone. For this, make an .apk file of the app. Note that you have three options for setting up live testing of the app, as given below.

1. If you have a wireless Internet connection on your PC and Android device, you can start building the app using MIT App Inventor without downloading any software to your computer. This option is used in this article, as explained above. For working offline, you need App Inventor Companion app on your device and App Inventor 2 Setup on your PC.
2. If you do not have a wireless Internet connection, install the software on your computer so that you can connect to your Android device over USB.
3. If you do not have an Android device but PC is available, use Emulator on your PC.

Note: The source code (.aia) and app (.apk) of this project are available for free download at *source.efymag.com.*

On the top of MIT App Inventor, there is Build option. Click on it and you will have two options to get .apk file. Select any option, you will get the app in .apk form. Download .apk and install it on your smartphone.

Testing Android app with Bluetooth module

To test the app, make the circuit with Bluetooth module (HC-05), as shown in Figure 20. Upload the source code (HC05_Tested.ino) to Arduino Uno using a USB cable and Arduino IDE. The source code is given below.

```
int led=13;
void setup() {
 pinMode(led,OUTPUT);
 Serial.begin(9600);
}

void loop() {
 if(Serial.available()>0){
  int a = Serial.read();
  if(a=='1'){
    digitalWrite(led,HIGH);
    Serial.println("Led is On");
  }
  else if(a=='0'){
    digitalWrite(led,LOW);
    Serial.println("Led is Off");
  }
 }
}
```

Connect HC-05 Bluetooth module with your Android's Bluetooth (from setting). Next, open HC05_with_phone app, connect Bluetooth by clicking on Available Bluetooth device and select HC-05. You will see Connected message on your cellphone. You can now turn on or turn off the LED (LED1). Status of the LED will also be shown on the app. END

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